Exam 3 Content Overview:

- **Line integrals** include the Fundamental Theorem of Line Integrals for conservative vector fields and Green's Theorem.
- **Surface integrals** include parameterization of surfaces, surface integrals of scalar functions, and surface integrals of vector fields (i.e., flux integrals).
- **Stokes' Theorem** connects the two: it equates the line integral (circulation) of $\mathbf{F}$ around a closed curve C to the surface integral (flux) of $\operatorname{curl}\mathbf{F}$ through any oriented surface S with boundary C.

Line Integrals:

1. **Fundamental Theorem of Line Integrals (conservative vector fields)**

   If the vector field $\mathbf{F} = P(x,y,z)\,\mathbf{i} + Q(x,y,z)\,\mathbf{j} + R(x,y,z)\,\mathbf{k}$ is **conservative**, i.e., $\mathbf{F} = \nabla f$, then the line integral of $\mathbf{F}$ along **any curve** C from A to B is

   $$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A).$$

   The line integral is path-independent, and $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C.

   Procedure:

   Step 1: Verify $\mathbf{F}$ is conservative by checking $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$.

   Step 2: Find a potential function f such that $\mathbf{F} = \nabla f$.

   Step 3: Plug initial and terminal points, A and B, of the curve into f.

   Practice: Q1–5

2. **Green's Theorem**

   Let R be a region in the xy-plane, and let C be its boundary, a simple **closed** curve traversed counterclockwise (positive orientation). For a **2D** vector field $\mathbf{F} = P(x,y)\,\mathbf{i} + Q(x,y)\,\mathbf{j}$:

   - Circulation (tangential) form:

     $$\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

     Compute the double integral using either rectangular or polar coordinates, whichever is easier.

   - Flux (normal) form:

     $$\oint_C \mathbf{F} \cdot \mathbf{N}\, ds = \iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA,$$

     where $\mathbf{N}$ is the outward unit normal to C.

Practice: Q6–13

Surface Integrals:

1. **Parameterization of Surfaces**

   Find parameterization equations for surfaces (sphere, cone, cylinder, plane, surface of revolution, circular disk, general surfaces).

   Practice: Q36

2. **Surface integrals of scalar functions**

   ✓ To set up $\iint_S f(x,y,z)\,d\sigma$:

   Step 1: Find a parameterization for the surface, $\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$.

   Step 2: Calculate $\mathbf{r}_u \times \mathbf{r}_v$.

   Step 3: Calculate the magnitude $|\mathbf{r}_u \times \mathbf{r}_v|$.

   Step 4:

   $$\iint_S f(x,y,z)\,d\sigma = \iint_D f\big(x(u,v), y(u,v), z(u,v)\big)\,|\mathbf{r}_u \times \mathbf{r}_v|\,du\,dv,$$

   where D is the parameter domain in the uv-plane.

   ✓ Applications: (1) Surface area: take $f = 1$, so Area $= \iint_S 1\,d\sigma$. (2) Mass of a surface with density $\rho(x,y,z)$: Mass $= \iint_S \rho\,d\sigma$.

   Practice: Q14–22

3. **Surface integrals of vector fields** (Flux Integrals)

   To compute $\iint_S \mathbf{F}(x,y,z)\cdot\mathbf{N}\,d\sigma$:

   Step 1: Find a parameterization for the surface, $\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$.

   Step 2: Calculate $\mathbf{r}_u \times \mathbf{r}_v$.

   Step 3: Check whether the normal vector $\mathbf{r}_u \times \mathbf{r}_v$ agrees with the prescribed orientation of S (e.g., upward or outward unit normal $\mathbf{N}$). If not, reverse its sign.

   Step 4:

   $$\iint_S \mathbf{F}(x,y,z)\cdot\mathbf{N}\,d\sigma = \iint_D \mathbf{F}\big(x(u,v), y(u,v), z(u,v)\big)\cdot\mathbf{N}\,|\mathbf{r}_u \times \mathbf{r}_v|\,du\,dv,$$

   where D is the parameter domain.

Step 5: Evaluate the double integral with respect to u and v.

Practice: Q23–28

Stokes' Theorem (connecting line and surface integrals)

Stokes' Theorem: For **3D** vector fields $\mathbf{F} = P(x,y,z)\,\mathbf{i} + Q(x,y,z)\,\mathbf{j} + R(x,y,z)\,\mathbf{k}$, and a piecewise-smooth, simple **closed** curve $C = \partial S$ bounding an oriented surface S, where C carries the induced (positive) orientation, consistent with $\mathbf{N}$ by the right-hand rule:

$$\oint_C \mathbf{F}\cdot\mathbf{T}\,ds = \iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$$

Practice: Q29–35.

Exam 3 focus: convert the flux integral of $\operatorname{curl}\mathbf{F}$ across S into the circulation of $\mathbf{F}$ around its boundary curve $C = \partial S$, i.e., evaluate $\iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$ by computing $\oint_C \mathbf{F}\cdot d\mathbf{r}$.

Final exam: the reverse direction will also be tested: evaluate a circulation $\oint_C \mathbf{F}\cdot d\mathbf{r}$ by computing $\iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$ over a suitable surface S bounded by C.

Conceptual Questions:

- Identify which compositions of gradient, curl, and divergence are well-defined, based on whether each operator acts on a scalar field or a vector field. "Select all that apply" **(Practice 37, HW9 Problem 19)**.
- The physical interpretation of curl (circulation density, i.e., local rotation) and divergence (flux density per unit volume: a source if $\operatorname{div}\mathbf{F} > 0$, a sink if $\operatorname{div}\mathbf{F} < 0$). **(Practice 38–39)**
- Choose among the Fundamental Theorem of Line Integrals, Green's Theorem, Stokes' Theorem, and the direct parameterization method: select all methods that could evaluate a given line integral. **(Practice 40–41; see the decision guide below.)**

Choosing a method for a line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$:

- $\mathbf{F}$ is **conservative** ($\operatorname{curl}\mathbf{F} = \mathbf{0}$) $\Rightarrow$ **Fundamental Theorem of Line Integrals**: $\int_C \mathbf{F}\cdot d\mathbf{r} = f(B) - f(A)$, for **any curve** C (open or closed; the integral is 0 if C is closed).
- C is **closed** and $\mathbf{F}$ is a **2D** field $\Rightarrow$ **Green's Theorem**: convert to a double integral over the enclosed region R.
- C is **closed** and $\mathbf{F}$ is a **3D** field $\Rightarrow$ **Stokes' Theorem**: convert to the flux of $\operatorname{curl}\mathbf{F}$ through a surface S with $\partial S = C$.
- **Any** $\mathbf{F}$, **any curve** C $\Rightarrow$ **Direct parameterization**: $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\mathbf{r}'(t)\,dt$. This method always works; the theorems above are shortcuts. It is the only option when $\mathbf{F}$ is not conservative and C is **not closed**.